

Measuring consistency between individuals of affective responses to ASMR stimuli using intersubject correlation

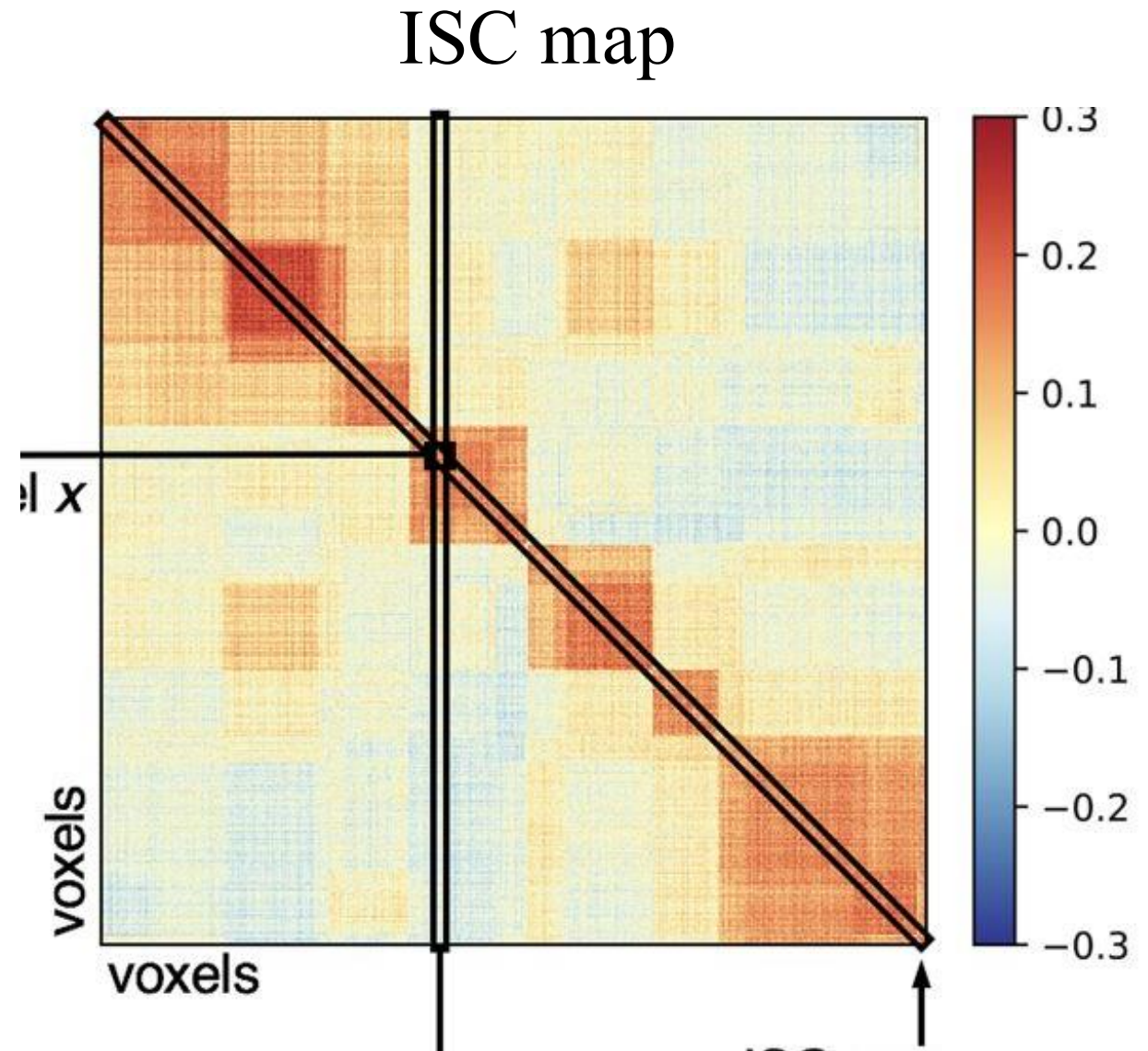
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Introduction

- Intersubject correlation (ISC)

By computing correlations across subjects ISC analyses are less susceptible to idiosyncratic physiological noise and head motion (Simony et al., 2016).



Nastase, S. A., Gazzola, V., Hasson, U., & Keysers, C. (2019)

Introduction

Intersubject correlation (ISC)

- Data-driven analysis
- Natural viewing setting

Traditional method (GLM)

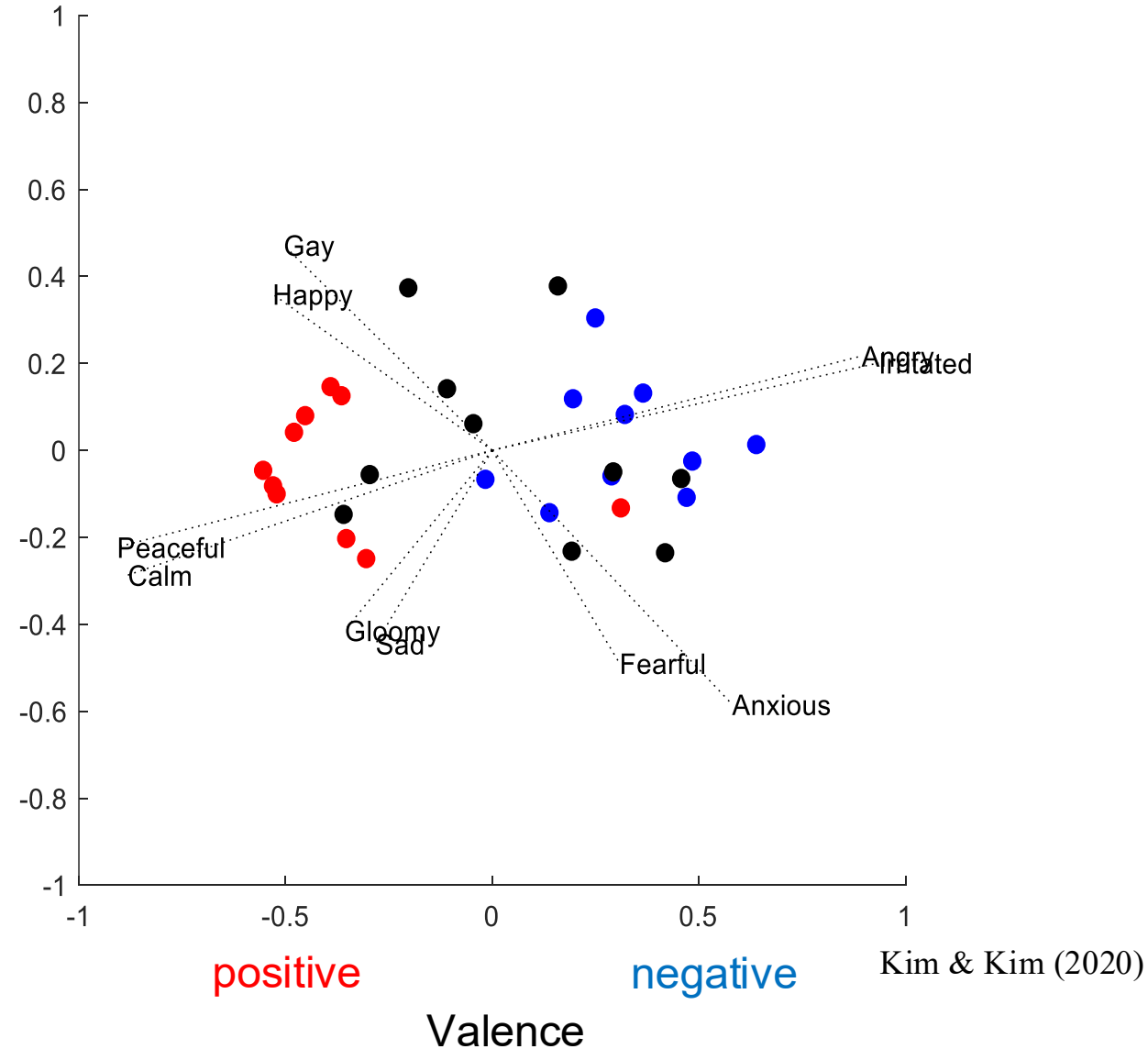
- Pre-defined model
- Controlled experimental setting

ISC analyses allow us to measure shared content across participants and experimental conditions (Nastase, Gazzola, Hasson, & Keysers, 2019).

Introduction

- Autonomous Sensory Meridian Responses (ASMR)

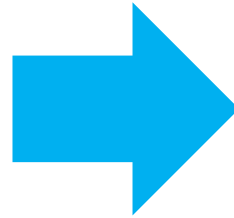
- ASMR triggers
- Kim and Kim (2020)'s finding of ASMR on affect
- Affect experience can be distinguished using ASMR stimulus



Research Purpose

- Previous research

- Limitedly used ISC to fMRI data (Najafi, Kinnison, & Pessoa, 2017)
- Valence dimension (Kim & Kim, 2020)
- Magnitude of emotion experience (Gao, Wedell, Kim, Weber, & Shinkareva, 2018)



- This study

- Application ISC to the behavioral data
- Valence and Modality dimensions
- Emotional experience pattern across individuals

Research Method

- Participant
 - 33 students (men: 6, women: 27)
- Stimulus
 - Kim and Kim (2020)'s the ASMR stimulus set (30) + Audiovisual set (30)
 - 10 adjective ratings
 - ✓ Fearful, Anxious, Gay, Peaceful, Happy, Sad, Irritated, Gloomy, Calm, Angry
 - Two factors design
 - ✓ Valence: (Negative vs Neutral vs Positive) and Modality: (Audio only vs Audiovisual)

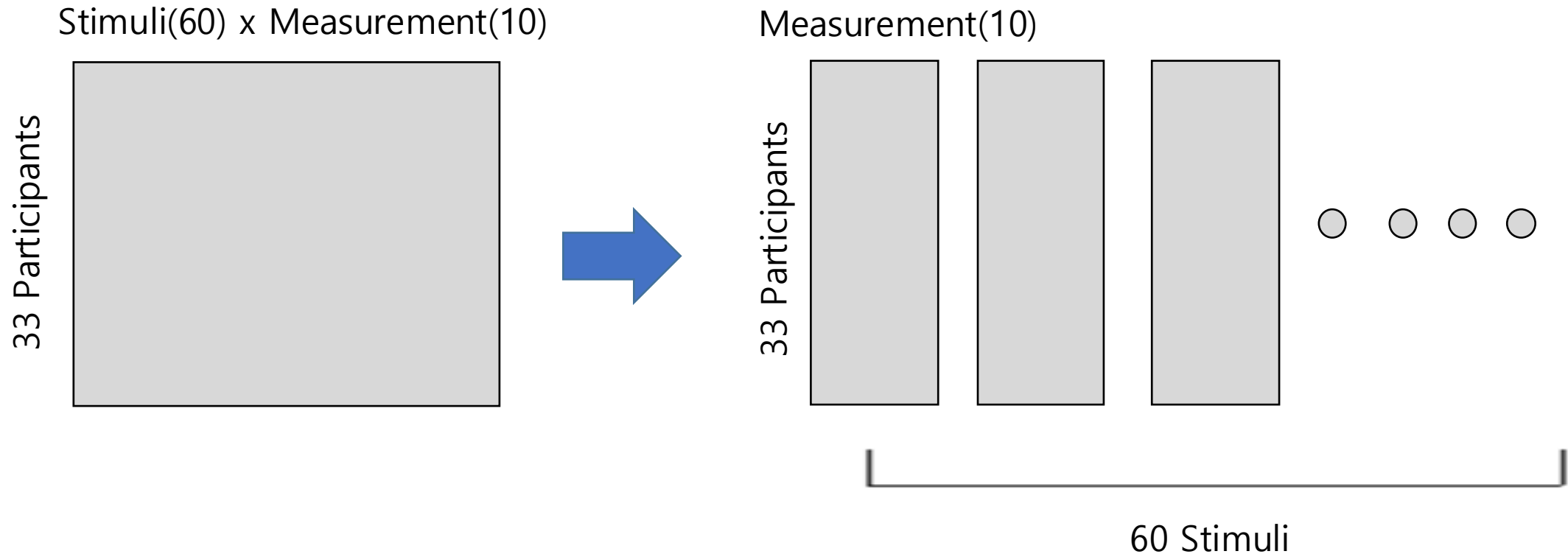
Research Method

- Procedure

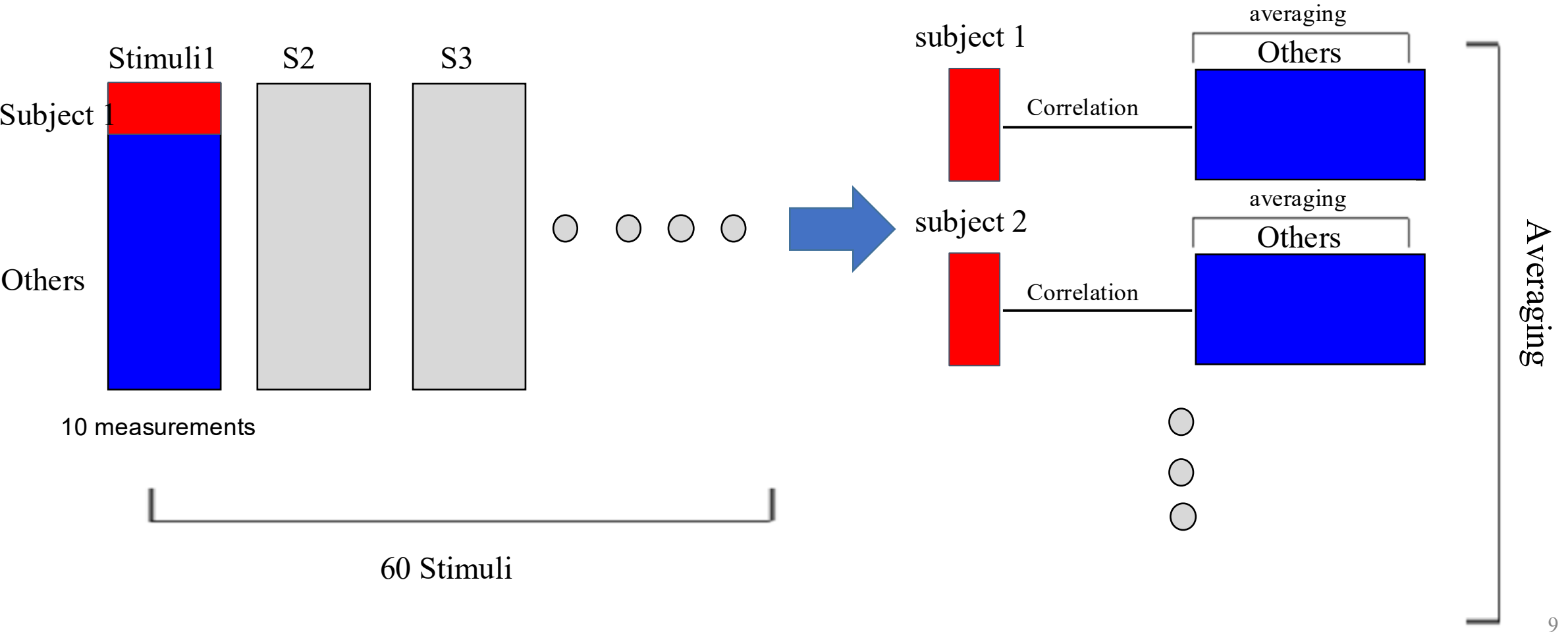
- Online experiment through Google docs
- Randomly ordered ASMR stimulus was presented for 30 second
- Rating to the ASMR stimulus
 - ✓ 5 Likert scale
 - ✓ 1; Strongly disagree, 2; Disagree, 3; Neither disagree nor agree, 4; Agree, 5; Strongly agree

Research Method

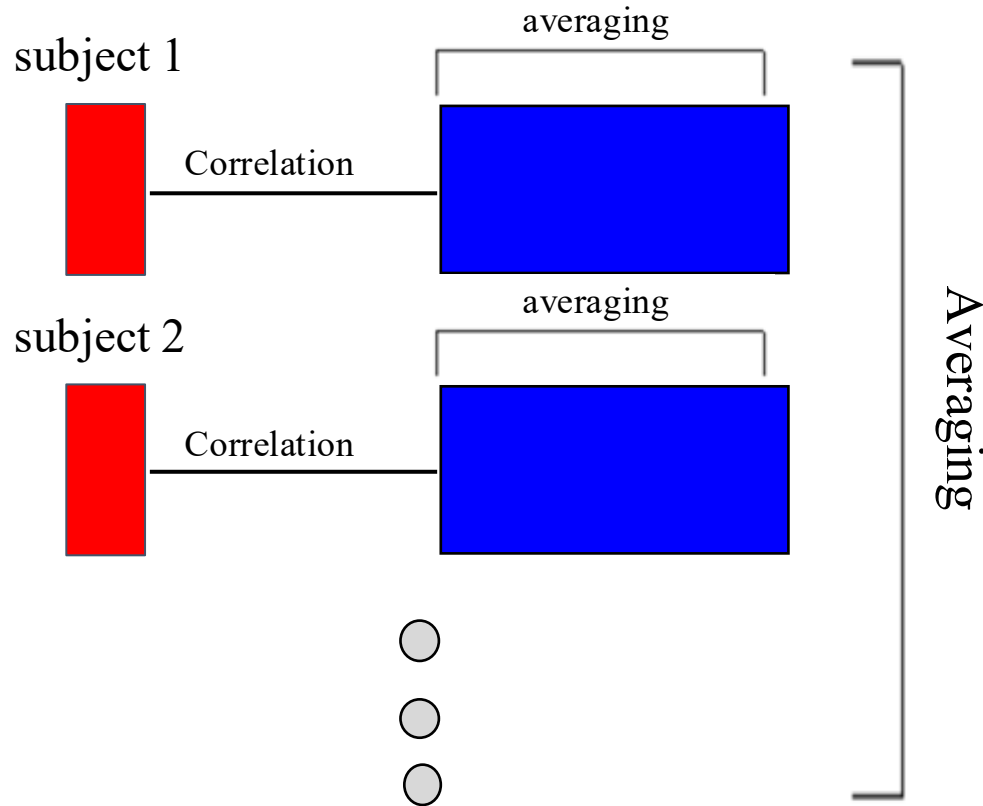
- Data processing and Analysis



Research Method



Research Method



Correlations between the same targets across subjects (Diagonal values)

Repeated measures ANOVA was used with the Diagonal value

Extend!

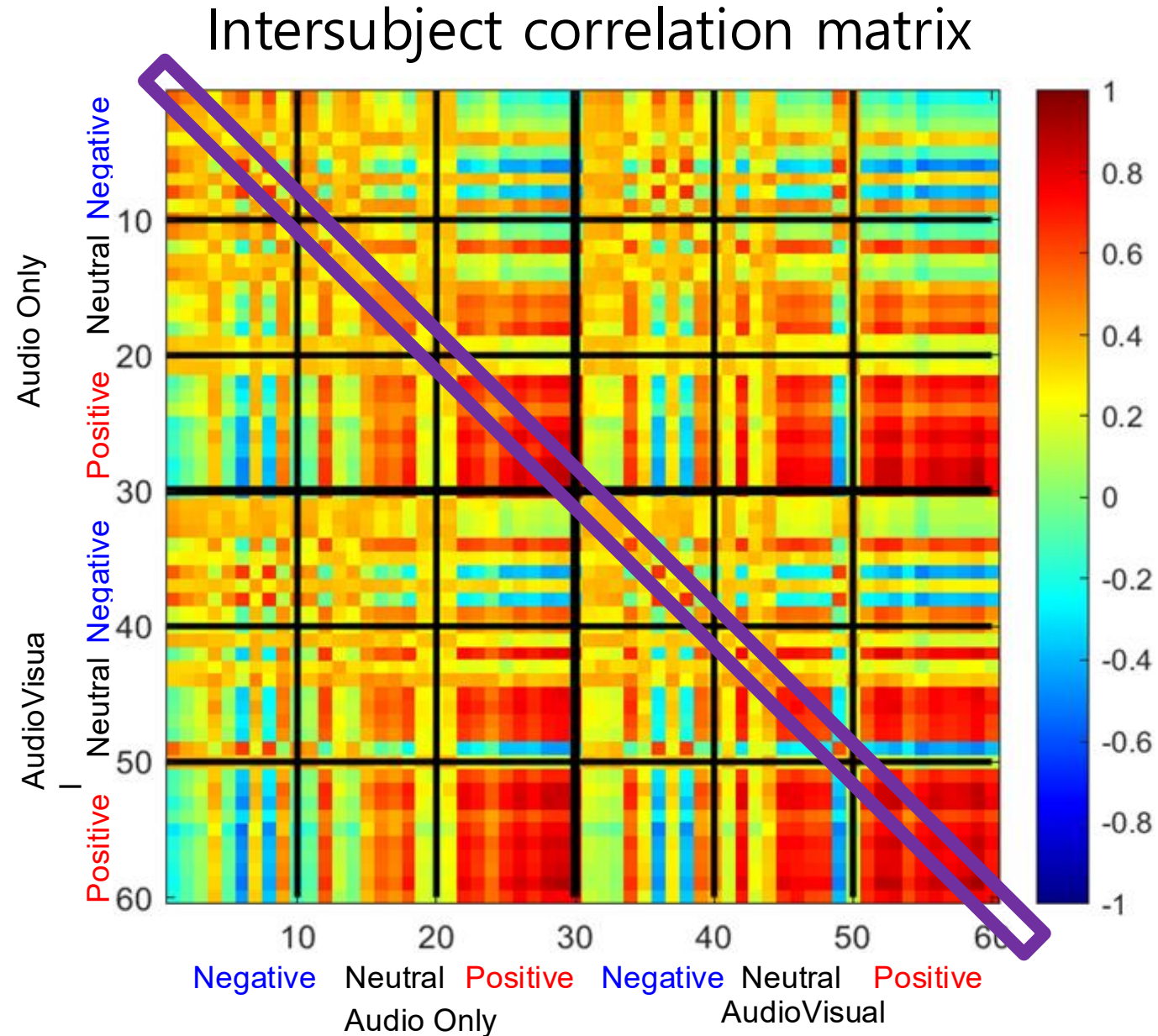
Correlations between all inter-stimuli across subjects (off-Diagonal values)

Intersubject correlation matrix was created

Result

- ISC map

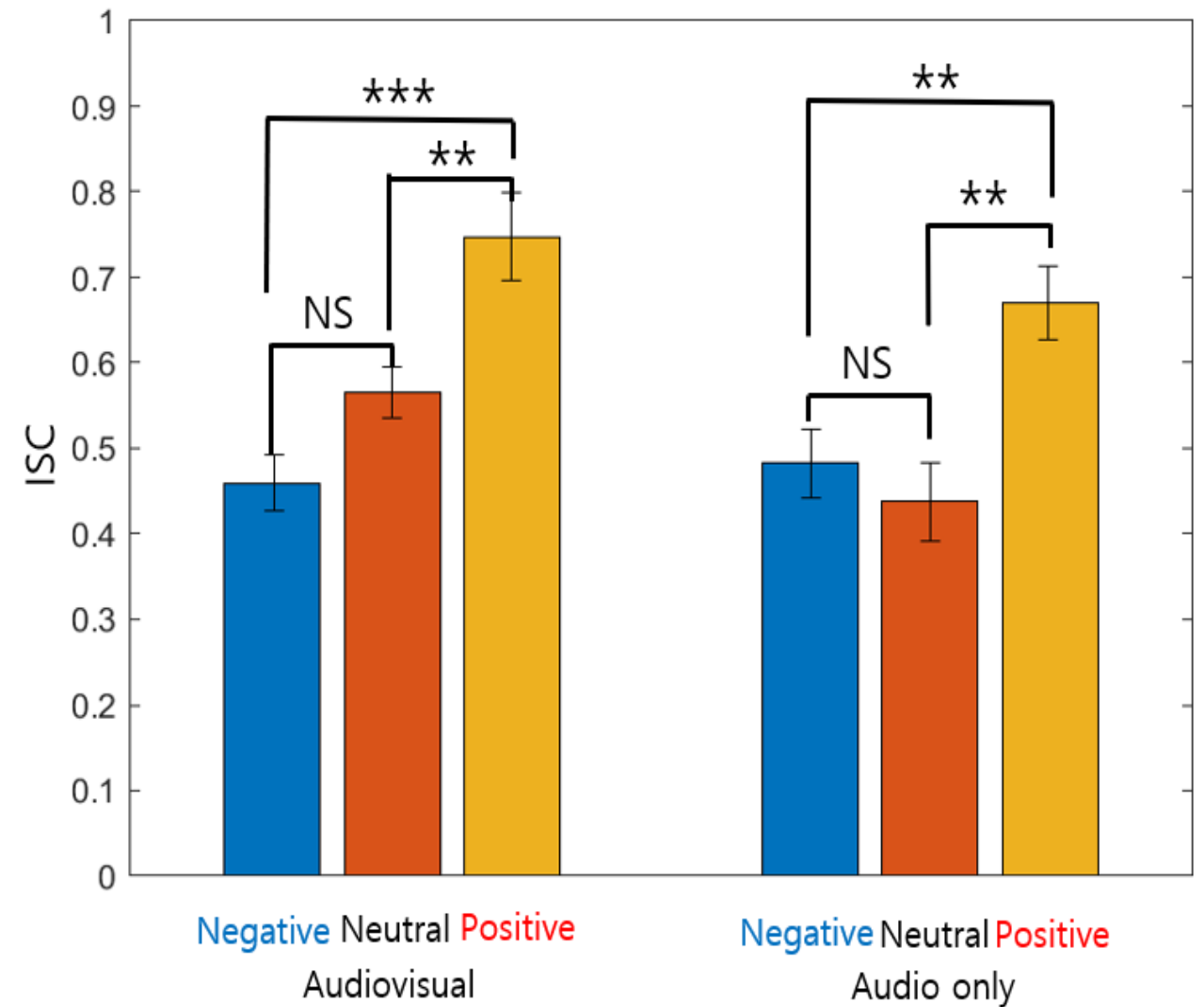
- The color bar indicates the range of ISC from -1 to 1
- Each of the x and y axis represents the 60 ASMR stimuli.
- The diagonal values reflect the correlations only computed between the same targets across subjects
- The off-diagonal values represent all inter-stimuli correlations across subjects.



Result

● Two-way ANOVA

- Diagonal correlations
- **Main effect of Valence**, $F(2,18) = 15.863$, $p < .001$, $\eta_p^2 = .652$ and **Modality**, $F(1,9) = 8.467$, $p = .017$, $\eta_p^2 = .485$
- **Interaction between Valence and Modality**, $F(2,18) = 4.840$, $p = .021$, $\eta_p^2 = .350$.
- Followed up repeated T test indicated **significant effects between modality only for neutral** ($t = -3.125$, $p < 0.05$) and **positive stimulus conditions** ($t = -2.688$, $p < 0.05$).



Discussion

- Different affect consistency to the distinct stimulus across subjects
 - Audiovisual > Auditory
 - Positive > Neutral = Negative
- Supporting arguments
 - Multimodal facilitation effect (Gao et al., 2018)
 - Emotion enhances the synchronization of physiological responses across participants (Bracken, et al., 2014)
- Counter findings
 - Nummenmaa et al. (2012) argued that negative emotions increased the synchronization of regional brain areas across participants.

Conclusion

- Application ISC to the behavioral data
 - ASMR affect ratings
 - Valence and Modality dimensions
 - Individual's different consistency with the ISC map and using Two-way repeated measures ANOVA.
- Limitation and Future Research
 - Online experiment
 - Future studies may include physiological data as additional features.

Reference

- Bracken, B. K., Alexander, V., Zak, P. J., Romero, V., & Barraza, J. A. (2014). *Physiological synchronization is associated with narrative emotionality and subsequent behavioral response*. Paper presented at the International conference on augmented cognition.
- Gao, C., Wedell, D. H., Kim, J., Weber, C. E., Shinkareva, S. V. J. C., & Emotion. (2018). Modelling audiovisual integration of affect from videos and music. *32*(3), 516-529.
- Hasson, U., Nir, Y., Levy, I., Fuhrmann, G., & Malach, R. J. s. (2004). Intersubject synchronization of cortical activity during natural vision. *303*(5664), 1634-1640.
- Kim, H., & Kim, J. (2020). Multidimensional scaling of emotional responses to ASMR sounds. *Annual fall conference of Korean Society for Emotion & Sensibility, 2020(-)*, 67-68.
- Kim, M., & Min, K. (2004). Emotional experience and emotion regulation in old age. *The Korean Journal of Psychology: General*, *23*(2), 1-21.
- Najafi, M., Kinnison, J., & Pessoa, L. (2017). Dynamics of Intersubject Brain Networks during Anxious Anticipation. *11*(552). doi:10.3389/fnhum.2017.00552
- Nastase, S. A., Gazzola, V., Hasson, U., & Keysers, C. (2019). Measuring shared responses across subjects using intersubject correlation. *Social Cognitive and Affective Neuroscience*, *14*(6), 667-685. doi:10.1093/scan/nsz037 %J Social Cognitive and Affective Neuroscience
- Nummenmaa, L., Glerean, E., Viinikainen, M., Jääskeläinen, I. P., Hari, R., & Sams, M. J. P. o. t. N. A. o. S. (2012). Emotions promote social interaction by synchronizing brain activity across individuals. *109*(24), 9599-9604.
- Paulmann, S., Pell, M. D. J. M., & Emotion. (2011). Is there an advantage for recognizing multi-modal emotional stimuli? , *35*(2), 192-201.
- Simony, E., Honey, C. J., Chen, J., Lositsky, O., Yeshurun, Y., Wiesel, A., & Hasson, U. (2016). Dynamic reconfiguration of the default mode network during narrative comprehension. *Nat Commun*, *7*, 12141. doi:10.1038/ncomms12141

Thank you



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